

Element A

Problem:

The Nashville Zoo seeks a device to hold the X-ray detection plate in their large animal barns.

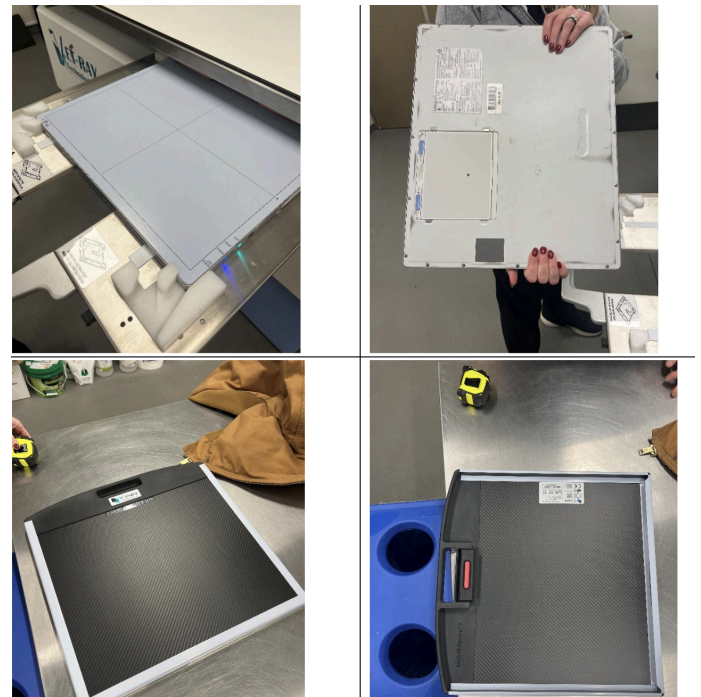
Currently, in order to take an X-ray of a giraffe's head, one person has to hold up the X-ray machine on one side of the animal while another person holds up the X-ray detection plate on the other side. However, it is difficult to raise and hold the detection plate up to the desired location while maintaining the correct rotation to take an accurate X-ray. As a result, the giraffe must sometimes be knocked out to safely take an X-ray,



which carries risks of injury or death. Thus, we are asked to create a device that can hold the X-ray detection plate in a desired placement. This would be able to maintain a constant position and orientation better than a person and completely replace said person during the X-ray, meaning X-rays could be taken with one less person.

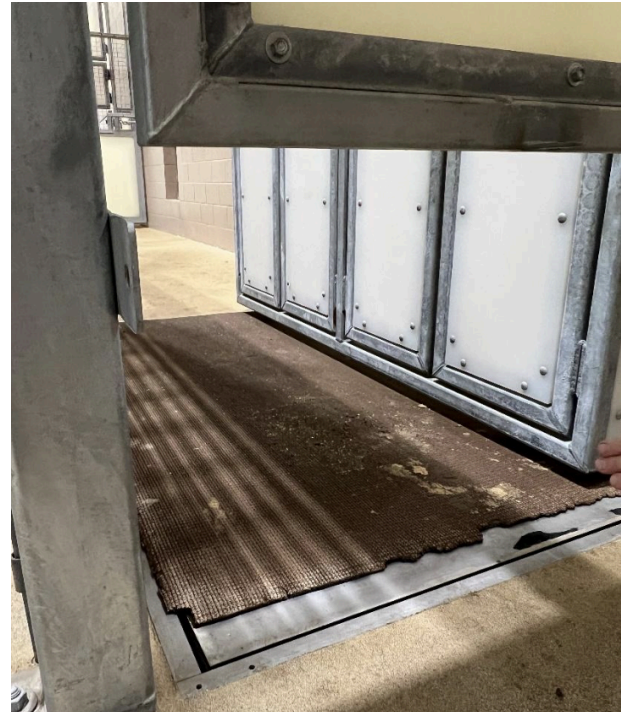
This plate holder should be adjustable in 3D space, both translational and rotational, to accommodate the different giraffes and the variety of X-rays. It may attach to the existing structural poles inside the giraffe enclosure, and if it does, it must be removable. It should also be able to withstand strong pushes from the giraffe without being significantly shifted or damaged.

Apart from being used for giraffe X-rays of their neck and head, the zoo would like a similar device



to hold their X-ray plates for okapi and giraffe leg X-rays.

Since the okapi X-rays are of the head area and are the most similar in nature to the giraffe head X-rays (wanting an adjustable mount attached to a pole), this will be a secondary objective. As for the case of the giraffe leg X-rays, that has separate criteria and constraints; thus, it will be a tertiary objective.



Stakeholders for all three objectives

Primary stakeholder A:

- The person who would've otherwise held the X-ray plate for the head/neck X-rays

Primary stakeholder B:

- The rest of the team taking the X-ray.

Primary stakeholder C:

- The group of animals getting the X-rays.

Secondary stakeholder A:

- The medical team/other personnel at the zoo.

Secondary stakeholder B:

- Patrons/visitors of the zoo.

